

# WalletVibe Codebase Metrics & Cost Analysis

Executive Engineering Audit, Token Trajectory & Firestore Storage Analysis

|                                                                                                              |                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <div>TOTAL CODEBASE LINES &amp; TOKENS</div> <div>16,798 LOC</div> <div>191,877 Base Tokens (46 Files)</div> | <div>EST. 15X AGENTIC API COST</div> <div>\$11.54 USD</div> <div>≈ ₹964 INR (15x Iteration Multiplier)</div> |
| <div>HUMAN ENG. EQUIVALENT</div> <div>672 Hours</div> <div>≈ \$33,600 USD Value Created</div>                | <div>FIRESTORE TIER LIMIT</div> <div>1.0 GB</div> <div>Firebase Spark Free Quota Active</div>                |

■ TOTAL OVERALL DEVELOPMENT COST SUMMARY:

- TOTAL AGENTIC DEVELOPMENT COST (15x Tokens): **\$11.54 USD (≈ ₹964 INR)**
- EQUIVALENT HUMAN SOFTWARE DEV COST (672 Hours): **\$33,600.00 USD (≈ ₹28,05,000 INR)**
- TOTAL ENGINEERING COST SAVINGS: **\$33,588.46 USD (99.97% Cost Savings via Agentic Automation)**

## 1. Firestore Database Storage & 90% Threshold Protocol

Firestore Spark Free Plan includes **1.0 GB (1,024 MB / ~1,073,741,824 Bytes)** of stored data.

- **Live Occupancy Inspection:** Firebase Console (console.firebase.google.com) → Project **WalletVibe** → **Firestore Database** → **Usage** tab.
- **What if Storage Reaches 90% (900 MB)?** Upgrading to the Firebase **Blaze Plan** removes the 1 GB cap. Additional storage beyond 1 GB costs only **\$0.18 per GB per month** (approx ₹15/month per extra GB).
- **Caching Protection:** WalletVibe uses local snapshot caching (ww\_cached\_snapshot\_\*) so clients sync delta changes without inflating Firestore reads or storage.

## 2. Agentic Cost & Token Trajectory Breakdown (15x Iteration Model)

**NOTE ON ITERATION MULTIPLIER:** To develop a 16,798 LOC production codebase, multiple agentic iterations happen (prompting, file reading, code generation, debugging & refactoring). The overall cost is calculated using a **15x iteration multiplier** ( $15 \times 191,877 = 2,878,155$  Total Tokens).

| Metric Category       | Token Volume   | Iteration Multiplier & Purpose    | Estimated Cost (USD) |
|-----------------------|----------------|-----------------------------------|----------------------|
| Raw Codebase Snapshot | 191,877 Tokens | 1.0x Base Repository (16,798 LOC) | Static Code Base     |

|                          |                  |                                 |                                       |
|--------------------------|------------------|---------------------------------|---------------------------------------|
| Agentic Input Trajectory | 2,302,524 Tokens | 12.0x Context & Workspace Reads | \$5.76 USD (@ \$2.50/1M)              |
| Agentic Output Tokens    | 575,631 Tokens   | 3.0x Generated Code & Refactors | \$5.76 USD (@ \$10.00/1M)             |
| TOTAL 15X AGENTIC COST   | 2,878,155 Tokens | 15.0x Full Development Cycle    | \$11.54 USD (≈ <span>₹964</span> INR) |

### 3. Top Codebase Files by Token Density

| File Path                               | Lines (LOC) | Characters    | Estimated Tokens                       |
|-----------------------------------------|-------------|---------------|----------------------------------------|
| src/styles.css                          | 2,333       | 60,982        | 16,482 tokens                          |
| src/components/ReportsView.jsx          | 1,664       | 86,588        | 23,403 tokens                          |
| src/components/BankSearchModal.jsx      | 1,040       | 49,792        | 13,457 tokens                          |
| src/components/AdminPanel.jsx           | 1,025       | 56,098        | 15,162 tokens                          |
| src/components/CsvImportModal.jsx       | 834         | 39,682        | 10,724 tokens                          |
| src/components/PersonMergeModal.jsx     | 683         | 28,965        | 7,828 tokens                           |
| src/App.jsx                             | 589         | 22,818        | 6,167 tokens                           |
| src/components/RatingModal.jsx          | 567         | 22,349        | 6,040 tokens                           |
| src/api/subscription.js                 | 550         | 18,947        | 5,121 tokens                           |
| src/components/LegalModal.jsx           | 495         | 28,249        | 7,637 tokens                           |
| src/components/SubscriptionModal.jsx    | 486         | 27,585        | 7,454 tokens                           |
| src/components/MultiSelectCombobox.jsx  | 416         | 15,416        | 4,166 tokens                           |
| src/components/SettingsModal.jsx        | 399         | 19,579        | 5,291 tokens                           |
| src/components/Header.jsx               | 392         | 18,041        | 4,879 tokens                           |
| GROSS TOTAL (Entire 46 Workspace Files) | 16,798 LOC  | 710,974 Chars | 191,877 Base Tokens (2.88M Trajectory) |